

Gabrielli Fertig de Oliveira

FullStack Developer

CONTACT

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EDUCATION

2022 - 2025: Federal Institute of Santa Catarina (IFSC)
Technical High School in Electronics

2026 - present: Federal University of Santa Catarina (UFSC)
Bachelor in Information Systems

SKILLS

- Specialty: Python + TypeScript + React/Next.js + Node.js + PostgreSQL + Docker + AI/Computer Vision
- Programming: Python, TypeScript, JavaScript, Java, C, C++, C#, PHP
- Frontend & Full-Stack: React, Next.js, Vite, SvelteKit, HTML5, CSS3, Tailwind CSS, Bootstrap, Three.js, jQuery, WordPress
- Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, Firebase, Mongoose
- Databases & Data: PostgreSQL, MySQL, MongoDB, SQL, relational database design, Python scripting, ETL and data processing
- AI & Computer Vision: Artificial Intelligence, Machine Learning, Computer Vision, OpenCV, YOLOv8, MediaPipe, Google Gemini, neural network training and deployment, prompt engineering
- Robotics & Autonomous Systems: ROS 2, ArduPilot, Pixhawk, MAVLink, WPILib, NVIDIA Jetson, Raspberry Pi, VSLAM, Visual Odometry, autonomous navigation
- Cloud, DevOps & Infrastructure: AWS, Firebase, Docker, Docker Compose, Linux terminal, Vercel, Git, GitHub
- Testing & Code Quality: Jest, React Testing Library, ESLint, Husky, Zod
- Architecture & Tools: Full-Stack Architecture, API Integration, Authentication, JWT, CI/CD, Agile Development

LANGUAGE

- Portuguese: Native
- English: Fluent
- Spanish: Intermediate

WHO AM I

I build things that move, think, and scale.

I'm an Information Systems student, software developer, and robotics researcher working at the intersection of AI, Computer Vision, full-stack development, and autonomous systems.

I've programmed robots that competed internationally in the United States, developed AI-powered autonomous drones, trained neural networks for real-world computer vision, and built full-stack products from the frontend all the way to cloud infrastructure and AI integrations.

I'm comfortable jumping from React and TypeScript to Python and Computer Vision, from PostgreSQL and Docker to ROS 2 and autonomous navigation. More importantly, I learn fast, take ownership, and enjoy solving problems where the answer isn't already written down.

I'm not looking for a role where I only write code. I want to build ambitious technology, learn whatever the problem demands, and take responsibility for making it work.

PROFESSIONAL EXPERIENCE

Research Scholarship Holder — Autonomous Drones & AI

IFSC — Research Project | Current

- Developing an autonomous drone system from scratch for electrical infrastructure inspection, combining full-stack development, artificial intelligence, computer vision, robotics, and embedded systems.
- Leading the development of a full-stack platform for drone operation, data processing, visualization, and system management using React, Vite, Node.js, Express, PostgreSQL, Python, FastAPI, Docker, and Docker Compose.
- Developing and integrating AI and Computer Vision models for autonomous perception and decision-making, including object detection, visual analysis, and navigation-related tasks.
- Working with ROS 2, ROS Humble, NVIDIA Jetson, Docker, VSLAM, and advanced visual odometry for autonomous robotic systems.
- Designing software architectures that connect the drone's onboard systems, computer vision pipeline, backend services, databases, and web interfaces into an integrated platform.
- Working across software and hardware layers, troubleshooting real-world integration issues and testing autonomous systems in laboratory environments.

Full-Stack & AI Developer — Actor Copilot

AI-powered Acting & Script Training Platform | 2026

- Developing an interactive AI application designed to help actors rehearse scripts through real-time speech recognition, conversational AI, and expressive text-to-speech.
- Building the application with Next.js 16, React 19, TypeScript, and Tailwind CSS, following a modern component-based architecture.
- Developing backend infrastructure with Firebase Authentication, Firestore, Cloud Functions, and Firebase Admin SDK, with deployment and hosting through Vercel.
- Integrating Google Gemini AI to power the Acting Coach and script-interaction pipeline, including AI-driven dialogue and acting assistance.
- Implementing document-processing workflows capable of handling PDF and DOCX scripts, allowing users to upload scripts and select characters/parts for interactive rehearsal.
- Integrating Stripe for payments, billing, and subscriptions, alongside Resend for transactional email and PostHog for product analytics.
- Building robust forms and data validation using React Hook Form and Zod, with authentication and security workflows involving JWT management and encryption.
- Implementing automated testing with Jest and React Testing Library, alongside ESLint and Husky for code quality and development workflows.
- Working across the entire product stack, from UI/UX and frontend architecture to APIs, databases, AI integration, authentication, payments, deployment, and analytics.

Drone Programmer — DronesIFSC Competition Team

IFSC | April 2024 – Present

- Developed autonomous programming for quadrotors competing in the Brazilian Robotics Competition (CBR), combining robotics, embedded systems, artificial intelligence, and computer vision.
- Implemented autonomous navigation and decision-making algorithms using Python, OpenCV, and computer vision techniques.
- Trained and deployed YOLOv8 neural networks for object detection and autonomous navigation, developing custom datasets and computer vision pipelines for competition environments.
- Developed computer vision algorithms using MediaPipe for drone navigation based on human gesture recognition.
- Worked with ArduPilot, Pixhawk, Raspberry Pi, MAVLink, and onboard computer systems to integrate autonomous software with drone hardware.
- Currently working with NVIDIA Jetson, ROS 2, ROS Humble, Docker, Docker Compose, VSLAM, and advanced visual odometry for autonomous robotic perception and navigation.
- Presented research on AI and Computer Vision for competitive drones as a guest speaker at the 2025 Olimpíada Baiana de Drones (OBAdrones), delivering a technical presentation alongside aerospace engineering professionals on the development and training of a neural network for autonomous drone applications.

Robotics Team Programmer & Technical Leader — FRC5800 Magic Island Robotics

2022 – 2024 | IFSC

- Programmed and competed with an FRC competition robot, developing autonomous and teleoperated systems in Java and WPILib and integrating software with complex mechanical and electronic systems.
- Took on leadership responsibilities within the programming division, coordinating robot software development and troubleshooting critical issues during international competitions.
- Served as both robot programmer and pilot/driver at three international FRC competitions in the United States, making real-time technical decisions and resolving software issues under high-pressure competition conditions.
- Contributed to the development of one of the team's first autonomous trajectory-programming systems, using tools including PathPlanner, odometry, AprilTags, Limelight, PhotonVision, and WPILib.
- Led the development of MagicScouting, a mobile scouting and data-analysis application built with SvelteKit, JavaScript, Tailwind CSS, and database integrations, improving match data collection and supporting strategic decision-making.
- Worked in a team of three to design and implement the scouting platform, including data collection, offline/local storage, wireless data transfer, and visualization workflows.
- Presented robotics projects and technical work at major technology events, including GitTogether Brasil, Latinoware 2023, and Build Season 2023 — Brasília.
- Participated in international robotics competitions, including the Brunswick Eruption Off Season in New Jersey, USA, where the team received a merit-based scholarship opportunity.

Web Development Intern — VegomTec

July 2023 – December 2023

- Maintained and improved a large-scale company website containing hundreds of pages, implementing frontend corrections, responsive adaptations, and usability improvements.
- Developed and maintained the frontend of a mobile business management application using HTML, CSS, JavaScript, and Bootstrap.
- Worked directly with an existing production codebase, identifying frontend issues and implementing practical solutions to improve the user experience and service-request workflow.
- Collaborated with the development team to adapt interfaces to different screen sizes and business requirements.

Research & Extension Researcher — IFSC

July 2023 – November 2024

- Conducted research in artificial intelligence, computer vision, autonomous robotics, and competition robots, applying Python and machine learning techniques to real-world robotic systems.
- Participated in the development of autonomous robot capabilities through computer vision, sensor integration, navigation algorithms, and AI-based decision-making.
- Developed and launched an Educational Robotics Platform using WordPress, designed to help public-school teachers introduce robotics without requiring previous robotics expertise or expensive equipment.
- Contributed to technology outreach and digital inclusion initiatives through robotics education for children and teenagers in underserved communities.
- Taught introductory robotics, Arduino, electronics, and programming concepts through workshops developed in partnership with Instituto Padre Vilson Groh (IVG) and IFSC.
- Participated in robotics outreach activities at the Hospital Infantil Joana de Gusmão, introducing children to robotics and technology through accessible educational activities.

Served as a robotics competition judge and participated in STEM education initiatives aimed at increasing access to technology and engineering education.

Electronics & Embedded Systems Researcher — IFSC

Research Project | 2024 – 2025

- Designed, implemented, and bench-tested electronic circuits for a medical-oriented device developed to measure electrical and sensor parameters in hospital environments.
- Programmed microcontrollers responsible for data acquisition, processing, and communication with external systems.
- Developed software and hardware integration workflows connecting embedded devices with cloud infrastructure.
- Developed an interactive dashboard for visualizing collected data, integrating Amazon Web Services (AWS) for cloud-based data storage and availability.
- Worked across electronics, embedded programming, data processing, web development, and cloud infrastructure, contributing to the complete development cycle of the research prototype.